

# N100 Financial Intelligence Platform

## Comprehensive Analyst & Operational Guide

### Executive Summary & Platform Scope

The N100 Financial Intelligence Platform is a state-of-the-art financial analysis, peer benchmarking, and screening system engineered for institutional equity analysts, portfolio managers, and risk research teams. The platform ingests multi-year financial statements across Nifty 100 constituents, computes over 30 financial ratios, and delivers real-time screener capabilities, interactive visual analytics, and automated PDF tearsheets.

#### Target Users

- **Equity Research Analysts:** Perform deep-dive financial profile assessments and ratio analysis.
- **Portfolio Managers:** Evaluate sector allocations, peer rankings, and risk metrics.
- **Risk & Credit Officers:** Monitor financial distress alerts, leverage thresholds, and cash-flow health.

#### Major Platform Capabilities

- **Automated ETL Engine:** Parses Excel statements into a normalized SQLite database with high data integrity.
- **Screener & Peer Benchmarking:** Real-time filtering across 20+ ratio metrics and 13 peer groups.
- **FastAPI REST Server:** High-performance JSON endpoints for automated integrations and data feeds.
- **Interactive Streamlit Dashboard:** 8 modular visualization screens with Plotly charts.
- **NLP & ML Engine:** Automated pros/cons text extraction and K-means company clustering.
- **ReportLab Tearsheet Generator:** Single-click institutional-grade PDF summary exports.

| Technology Layer       | Components & Libraries                    |
|------------------------|-------------------------------------------|
| Core Language          | Python 3.10+                              |
| Data Processing        | Pandas, NumPy, OpenPyXL                   |
| Database Backend       | SQLite 3.25+                              |
| API Framework          | FastAPI, Uvicorn, Pydantic v2             |
| Frontend UI            | Streamlit 1.42+, Plotly 6.0+              |
| Machine Learning / NLP | Scikit-Learn, SciPy, Custom Pattern Rules |
| Reporting & PDF        | ReportLab 5.0+, PyPDF                     |
| Testing & Quality      | Pytest, Black, Ruff                       |

# Getting Started & Environment Setup

Setting up the N100 Financial Intelligence Platform requires Python 3.10+ and a standard virtual environment. The repository relies on standard directory conventions and SQLite for zero-configuration database deployment.

## 1. Environment Prerequisites

Ensure Python 3.10 or later is installed on your operating system (Windows, macOS, or Linux).

```
python --version
```

## 2. Virtual Environment Creation

Create and activate an isolated Python virtual environment:

```
python -m venv venv
# Windows:
.\venv\Scripts\activate
# macOS/Linux:
source venv/bin/activate
```

## 3. Dependency Installation

Install required packages specified in the environment:

```
pip install -r requirements-dashboard.txt
```

## 4. Project Directory Hierarchy

```
N100_Financial_Intelligence_Platform/
■■■■ data/ # Source Excel financial files
■■■■ docs/ # Platform documentation & Analyst Guide PDF
■■■■ output/ # Exported CSVs, Excel reports & final database
■ ■■■■ NIFTY_SMALL_100.db # Authoritative SQLite Database
■ ■■■■ final_deliverables/ # Consolidated 23 project deliverables
■■■■ reports/ # Data quality reports & PNG figures
■■■■ src/ # Core Python package
■ ■■■■ analytics/ # Valuation, Peer, Trend & Cashflow engines
■ ■■■■ api/ # FastAPI server & REST router endpoints
■ ■■■■ dashboard/ # Streamlit app.py & modular page views
■ ■■■■ etl/ # Extraction, normalization & SQLite loading
■ ■■■■ kpi_engine/ # 30+ financial ratio calculation engine
■ ■■■■ nlp/ # Sentiment & pros/cons text parser
■■■■ tests/ # Pytest unit, integration & performance suite
```

# Running the ETL Pipeline

The N100 Data Engine ingests unstructured multi-year Excel balance sheets, profit & loss accounts, and cash flow statements. It normalizes financial metrics across divergent accounting line items and populates the SQLite database schema.

## ETL Process Workflow

- 1. **Extraction:** Scans data/ directory for raw company financial workbooks.
- 2. **Normalization:** Standardizes key line items (e.g. Total Revenue, EBITDA, PAT, Net Debt).
- 3. **Validation:** Asserts Balance Sheet equation (Assets = Liabilities + Equity) and handles missing values.
- 4. **KPI Calculation:** Calculates 30+ financial ratios across historical fiscal years.
- 5. **SQLite Loading:** Writes processed records into output/NIFTY\_SMALL\_100.db.

## Executing the Pipeline

Run the main ETL pipeline script directly from the repository root:

```
python run_etl.py
```

## ETL Output Verification

Upon successful completion, the pipeline outputs summary logs to console and updates the SQLite database. Detailed HTML data quality reports are automatically written to reports/.

| Database Table   | Description                               | Primary Key             |
|------------------|-------------------------------------------|-------------------------|
| companies        | Company metadata, sector, industry        | company_id              |
| financial_kpis   | Calculated annual ratios (ROE, ROCE, P/E) | id (company_id, period) |
| balance_sheet    | Normalized balance sheet items            | id (company_id, period) |
| income_statement | Normalized P&L items                      | id (company_id, period) |
| cash_flow        | Normalized cash flow items                | id (company_id, period) |
| peer_percentiles | Peer-relative metric percentiles          | id (company_id, metric) |

# Streamlit Dashboard Overview

The N100 Streamlit Dashboard provides a responsive, web-based analytics portal. It connects directly to the SQLite backend and leverages Streamlit caching for sub-second page transition times.

## Launching the Application

Execute the exact Streamlit entry point command from the repository root:

```
streamlit run src/dashboard/app.py
```

## Accessing the Portal

Once initialized, open your browser and navigate to:

**Local URL:** http://localhost:8501

**Network URL:** http://<your-ip>:8501

## Dashboard Architecture & Pages

The dashboard features a persistent sidebar with navigation controls, company filters, and global settings. The 8 main analytics screens are structured as follows:

| Page Number | Module Name     | Primary Analytics Function                                          |
|-------------|-----------------|---------------------------------------------------------------------|
| 01          | Home            | Platform Overview, System Health & Key Performance Cards            |
| 02          | Company Profile | Deep-dive Financial Statements, Ratio Grids & Historical Trends     |
| 03          | Screener        | Multi-criteria Filter Engine with Custom Sliders & Preset Scenarios |
| 04          | Peers           | Industry Benchmarking, Percentile Ranks & Radar Overlay             |
| 05          | Trends          | 12-Year CAGR Analysis, Multi-Metric Plots & Growth Tracking         |
| 06          | Sectors         | Sector Performance Heatmaps, Aggregations & Distribution Charts     |
| 07          | Capital         | Capital Allocation Distribution & Cash Flow Intelligence Grid       |
| 08          | Reports         | Valuation Flagging, PDF Tearsheet Generation & Batch Export         |

# Dashboard — Home & Company Profile

The Home and Company Profile screens provide the primary entry points for qualitative and quantitative stock analysis.

## 1. Home Screen Features

- **KPI Snapshot:** Displays total coverage metrics (92 Companies, 9 Sectors, 12 Fiscal Years).
- **System Health Check:** Live status indicator of database connection and loaded tables.
- **Quick Search:** Instant ticker look-up to auto-redirect to dedicated company profiles.

## 2. Company Profile Navigation

Select any company from the sidebar drop-down menu (e.g. TCS, RELIANCE, INFYS). The Profile view renders five core tab views:

- **Financial Overview:** Summary KPI cards including Market Cap, P/E Ratio, ROE, ROCE, and Debt/Equity.
- **Income Statement:** Multi-year breakdown of Revenue, Operating Profit, EBITDA, and Net Income.
- **Balance Sheet:** Assets, Equity, Non-Current Liabilities, and Working Capital trend lines.
- **Cash Flow Statement:** Operating Cash Flow (CFO), Capital Expenditure (CapEx), and Free Cash Flow (FCF).
- **Ratio Analytics Grid:** Complete 30+ ratio breakdown grouped by Profitability, Liquidity, Solvency, and Efficiency.

Analyst Interpretation Guidance

**Tip for Analysts:** Examine the FCF vs Net Profit relationship over a 3-year trailing window. Companies displaying strong Net Income growth without corresponding OCF generation may indicate aggressive revenue recognition or working capital deterioration.

## Dashboard — Screener & Peer Comparison

The Screener and Peer Comparison engines enable institutional investors to identify investment candidates and benchmark company performance against industry cohorts.

### 1. Stock Screener Usage

Navigate to **03\_screener** from the sidebar. Analysts can apply custom criteria sliders:

- **Profitability Filters:** Min Return on Equity (ROE  $\geq$  15%), Min ROCE ( $\geq$  18%).
- **Valuation Filters:** Max P/E Ratio ( $\leq$  25.0), Max P/B Ratio ( $\leq$  3.5).
- **Financial Health Filters:** Max Debt-to-Equity ( $\leq$  0.5), Min Interest Coverage ( $\geq$  4.0).
- **Growth Filters:** Min 3-Yr Revenue CAGR ( $\geq$  10%), Min 3-Yr Profit CAGR ( $\geq$  12%).

### Preset Screening Scenarios

Click preset scenario buttons for instant filter setup:

- **Quality Compounders:** High ROE ( $>20\%$ ), Low Debt (D/E  $<0.3$ ), Positive FCF.
- **Deep Value:** Low P/E ( $<15$ ), Low P/B ( $<1.5$ ), Solvent Balance Sheet.
- **High Growth:** Revenue CAGR  $>15\%$ , Profit CAGR  $>20\%$ .

### 2. Peer Comparison & Radar Analysis

Navigate to **04\_peers** to evaluate company positioning across its designated peer group (e.g. IT Services, Banking, Pharma):

- **Percentile Rankings:** Shows company percentile (0–100th percentile) across 10 key metrics.
- **Radar Overlay:** Generates a 10-axis polar plot comparing the company's percentile profile against the peer benchmark mean.

# Dashboard — Trends, Sectors & Capital Allocation

Macro analytics pages synthesize cross-sectional industry data to highlight secular growth patterns and capital deployment efficiency.

## 1. Trend Analysis (05\_trends)

Analyze 12-year historical financial trajectories (2012–2024). Select up to 4 financial metrics simultaneously to compare growth momentum and margin trends over time. Calculates compound annual growth rates (CAGR) automatically.

## 2. Sector Analytics (06\_sectors)

Aggregates metrics across 9 industry sectors (Information Technology, Banking & Financials, Pharmaceuticals, Consumer Goods, Industrial, Energy, Automobile, Materials, Telecom).

- **Sector Heatmaps:** Visualizes median ROE, Operating Margin, and Valuation Multiples across sectors.
- **Sector Bubble Chart:** Plots Sector ROE vs P/E Multiple with bubble size scaled by Total Market Cap.

## 3. Capital Allocation Intelligence (07\_capital)

Evaluates corporate cash deployment strategies over multi-year cycles. Categorizes cash utilization across CapEx (Reinvestment), Dividends/Buybacks (Shareholder Returns), Debt Reduction, and Cash Accumulation.

Cash Flow Matrix Categories

- **Reinvestment Champions:** CapEx / Operating Cash Flow > 60% with ROIC > 15%.
- **Shareholder Value Creators:** Dividend Yield + Buyback Yield > 4% with FCF Conversion > 80%.
- **Capital Consumptive:** Operating Cash Flow < CapEx over 3 consecutive fiscal years.

## Dashboard — Reports & Valuation

The Reports & Valuation page (**08\_reports**) serves as the primary output hub for institutional reporting, valuation anomaly detection, and PDF tearsheet generation.

### 1. Automated Valuation Engine

Computes absolute and peer-relative valuation metrics including Price-to-Earnings (P/E), Price-to-Book (P/B), and Enterprise Value to EBITDA (EV/EBITDA). Applies automated valuation flagging rules:

- **Overvalued Flag:**  $P/E > 1.5 \times \text{Peer Group Median}$  AND  $ROE < \text{Peer Group Median}$ .
- **Undervalued Flag:**  $P/E < 0.7 \times \text{Peer Group Median}$  AND  $ROE > \text{Peer Group Median}$ .
- **Fair Value:** Multiples within +/- 15% of peer median.

### 2. Generating PDF Tearsheets

Analysts can generate instant institutional PDF tearsheets directly from the Streamlit UI:

1. Select Target Company from the drop-down menu on page 08\_reports.
2. Click the **Generate PDF Tearsheet** button.
3. Preview the generated tearsheet online or download the compiled PDF file.

### 3. Export Capabilities

Supports one-click export of underlying datasets to Excel and CSV formats:

- output/valuation\_summary.xlsx: Complete valuation master spreadsheet.
- output/valuation\_flags.csv: Mispricing flags and anomaly alerts.



## FastAPI Server & REST API Reference

The N100 backend exposes a high-performance REST API built on FastAPI and Uvicorn. It features automatic Pydantic request validation, CORS middleware, OpenAPI specifications, and Swagger UI documentation.

### API Startup & Interactive Docs

Start the production FastAPI server using Uvicorn:

```
uvicorn src.api.main:app --reload --port 8000
```

**Base URL:** <http://localhost:8000/api/v1> | **Interactive Docs:** <http://localhost:8000/docs> | **OpenAPI JSON:** <http://localhost:8000/openapi.json>

### Authoritative Endpoint Curl Examples

#### 1. Health Endpoint

```
curl -X GET http://localhost:8000/api/v1/health
```

#### 2. Companies Endpoint

```
curl -X GET http://localhost:8000/api/v1/companies
```

#### 3. Specific Company (TCS)

```
curl -X GET http://localhost:8000/api/v1/companies/TCS
```

#### 4. Screener Endpoint

```
curl -X GET "http://localhost:8000/api/v1/screener?min_roe=15&max_pe=25"
```

#### 5. Sectors Endpoint

```
curl -X GET http://localhost:8000/api/v1/sectors
```

#### 6. Peers Endpoint

```
curl -X GET http://localhost:8000/api/v1/peers/TCS
```

#### 7. Valuation Endpoint

```
curl -X GET http://localhost:8000/api/v1/valuation/TCS
```

#### 8. Portfolio Endpoint

```
curl -X GET http://localhost:8000/api/v1/portfolio
```

## PDF Reports & Tearsheet Generation Engine

The Report Generation Engine (src/reports/) builds publication-grade PDF documents using ReportLab. It features dynamic table layout, vector chart embedding, and two-pass page numbering.

### Report Types & Output Directories

- **Company Tearsheets:** Single/multi-page summary of financial health, ratios, and peer ranks. Saved to reports/tearsheets/.
- **Peer Comparison Reports:** Detailed markdown & PDF peer comparison files. Saved to output/peer\_reports/.
- **Sector Analysis Reports:** Sector-wide financial summaries. Saved to reports/sector/.
- **Portfolio Analysis Report:** Aggregate portfolio risk and exposure breakdown. Saved to reports/portfolio/.

### Programmatic Batch Generation Command

Generate tearsheets for all companies programmatically:

```
python -m src.reports.tearsheet_generator --all
```

### Handling Skipped Companies

If a company lacks sufficient historical balance sheet data to compute required ratio metrics, the generator logs a diagnostic entry to output/skipped\_tearsheets.csv and continues batch execution without failing.

#### *PDF Quality Inspection Checklist*

1. Open compiled PDF in any standard reader (Adobe Acrobat, Chrome, Preview).
2. Verify clean header line and page numbers ('Page X of Y').
3. Assert no overlapping text boxes or table column truncations.
4. Confirm vector graphics and colors render crisply.

## Advanced Analytics & Intelligence Engines

Beyond traditional ratio analysis, the platform incorporates advanced mathematical models, NLP sentiment engines, and machine learning algorithms.

### 1. NLP Pros & Cons Generator (src/nlp/)

Parses quantitative financial statements and qualitative commentary to generate human-readable investment highlights and risk alerts. Outputs consolidated findings to `output/pros_cons_generated.csv`.

### 2. Unsupervised ML Clustering (src/analytics/)

Applies K-Means clustering across standardized financial metrics (ROE, Leverage, Margin, Asset Turnover) to group companies into behavioral peer clusters independent of broad sector classifications. Outputs cluster labels to `output/cluster_labels.csv` and centroids to `output/cluster_profiles.csv`.

### 3. Financial Distress & Early Warning Alert System

Evaluates Altman Z-Score equivalents, Interest Coverage ratios, and CFO-to-Debt metrics to flag potential solvency distress. Flags are written to `output/distress_alerts.csv`.

### 4. Ratio Correlation Analysis

Computes cross-metric Pearson correlation matrices across all constituent companies. Visual correlation heatmaps are generated and saved to `reports/correlation_heatmap.png`.

## Troubleshooting & Operational FAQ

This section details diagnostic steps and practical shell commands for resolving common environment, database, and service issues.

| Symptom / Error             | Root Cause                                | Practical Solution Command                                               |
|-----------------------------|-------------------------------------------|--------------------------------------------------------------------------|
| Database Not Found          | SQLite file missing at expected path      | Run <code>python run_etl.py</code> to rebuild database.                  |
| Streamlit Entry Point Error | Invoking wrong path like app.py           | Run command: <code>streamlit run src/dashboard/app.py</code> .           |
| Port 8501 Already in Use    | Previous Streamlit instance active        | Kill process on 8501 or use <code>--server.port 8502</code> .            |
| Port 8000 Already in Use    | FastAPI server running in background      | Kill process on port 8000.                                               |
| API Unavailable / 404       | FastAPI server not started                | Launch API: <code>uvicorn src.api.main:app --reload --port 8000</code> . |
| Missing Python Module       | Virtual environment missing package       | Run <code>pip install -r requirements-dashboard.txt</code> .             |
| Empty Dashboard Charts      | No company selected or DB locked          | Select company from sidebar or verify DB path.                           |
| PDF Generation Failure      | ReportLab missing or directory unwritable | Ensure ReportLab installed and directory exists.                         |

## Quality Assurance & Testing Suite

The platform includes a robust Pytest suite covering unit calculations, API endpoint validation, NLP rule accuracy, report generation integrity, and load performance.

### Executing Test Commands

#### 1. Run Full Regression Suite

```
python -m pytest tests/ -q
```

#### 2. Run API Endpoint Tests

```
python -m pytest tests/api/ -q
```

#### 3. Run Analytics & KPI Engine Tests

```
python -m pytest tests/analytics/ -q
```

#### 4. Run NLP Engine Tests

```
python -m pytest tests/nlp/ -q
```

#### 5. Run Reports & Tearsheet Tests

```
python -m pytest tests/reports/ -q
```

#### 6. Run Performance & Load Benchmarks

```
python -m pytest tests/performance/ -q
```

#### 7. Run Integration End-to-End Suite

```
python -m pytest tests/integration/ -q
```

# Recommended Analyst Operational Workflow

Follow this 12-step standardized operational workflow for daily financial research, screening, and report compilation:

| Step    | Action Item             | Operational Instructions                                               |
|---------|-------------------------|------------------------------------------------------------------------|
| Step 1  | Verify Database         | Check that output/NIFTY_SMALL_100.db exists and is populated.          |
| Step 2  | Validate Data Quality   | Run python run_etl.py if raw Excel files have updated.                 |
| Step 3  | Start FastAPI Server    | Launch backend: uvicorn src.api.main:app --reload --port 8000.         |
| Step 4  | Launch Dashboard UI     | Start frontend: streamlit run src/dashboard/app.py.                    |
| Step 5  | Select Target Company   | Use sidebar company selector on Page 02_profile to view statements.    |
| Step 6  | Analyze Key Ratios      | Review Profitability, Solvency, and Free Cash Flow metrics.            |
| Step 7  | Run Custom Screener     | Navigate to Page 03_screener to filter candidates matching thesis.     |
| Step 8  | Benchmark Peer Group    | Navigate to Page 04_peers to inspect percentile ranks and radar plots. |
| Step 9  | Evaluate Sector Context | Navigate to Page 06_sectors to review sector margin heatmaps.          |
| Step 10 | Review Cash Deployment  | Navigate to Page 07_capital to analyze CapEx vs Dividend ratios.       |
| Step 11 | Inspect Valuation Flags | Navigate to Page 08_reports to check automated P/E mispricing flags.   |
| Step 12 | Export Tearsheet PDF    | Generate and download institutional PDF tearsheet for presentations.   |